

Bangladesh University of Engineering and Technology

Department of Computer Science and Engineering

CSE 108: Object Oriented Programming

Offline Assignment # 1 (Week 2)

Due Date:	12 December 2025 at 11:55pm
Cut-off Date:	12 December 2025 at 11:55pm
Full Marks:	20
Submission Method	WebFace (i.e., Moodle) Assignment System
Submitted File	Solutions to the given problems must be submitted in one file. The file name must be Ass1_YourStdID_Dec25.cpp Please don't use any other file name while submitting your assignment solutions through WebFace.
Reference:	Encapsulation – Data Hiding

Submission Policy

Assignment Submission Methods

- Assignments must be submitted through the WebFace (i.e., Moodle) Assignment Submission System. No submission based on any other channel like hardcopy, email is not allowed.

Late submission of assignments

- There is no **probation** of late submission of the assignment.

Extensions

- No request of date extension for submitting the assignment is permitted.

Assessment Policy

- You must ensure that others do not obtain access to your solutions for the purpose of copying a part of them. Where such plagiarism is detected, both of the assignments involved will be marked zero.
- Your assignment will be marked zero if you have posted any clues to assignment solutions before you receive your marked assignment back.
- Your assignment must not be AI generated. In such cases, your obtained marks in the assignment will be zero.

Assignment Problem:

A geometric shape like **line, triangle, rectangle**, etc., can be represented by a matrix. There are two end points for a line, three end-points for a triangle and four end-points for a rectangle. Each row of the matrix represents a point, and each column represents a dimension of it. For example, a triangle in two-dimensional (2D) space is represented as-

$$\begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \end{pmatrix}$$

While a triangle in three-dimensional (3D) space is represented as-

$$\begin{pmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{pmatrix}$$

The class “Figure” is written as a template to store information about a line, triangle or rectangle. The class has the following variables:

- The variable **row** is an integer type data that represents number of points in the Figure.
- The variable **col** is an integer type data that represents the dimension of the Figure space.
- The variable **name** is a string type data that represents the name of the Figure like “Line2D”, “Triangle3D”, “Rectangle2D”, etc.
- The variable **matrix** is a **2D array** that stores the points of the Figure.

Write down C++ codes for the Figure class that includes the following properties:

1. The **encapsulation properties** of object-oriented programming. (3)

2. Three **overloaded constructors** based on following requirements: $(1 + 4 + 2 = 6)$
 - (a) The default constructor creates a zero-size matrix (i.e., **row** = **col** = 0), zero-length name and initializes all elements of the matrix to zero.
 - (b) The parameterized constructor takes **two integers r and c** as well as a 1D integer array to fill up the matrix. Please note that the matrix

$$\begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \end{pmatrix}$$
 can be represented by a 1D array as $(x_1, y_1, x_2, y_2, x_3, y_3)$.
 The object name will be selected automatically by the constructor from the value of **row** and **col** of the object.
 - (c) The clone constructor that creates a clone of the existing object.
3. The **destructor method** changes the values of **row**, **col** and all elements of the matrix to zero as well as display a message that the object has been destroyed (say the message is like, "**Triangle2D has been destroyed**"). (3)
4. Write down two **overloaded methods** named **getSum()** and **getSum(int x, int y)**. The **getSum()** method returns the sum of all values of the matrix; while the **getSum(int m, int n)** method returns the sum of all values of the matrix with row number ranges from 0 to $m-1$ and column number ranges from 0 to $n-1$. The values of m and n must be less than the values of the **row** and the **col**; otherwise, it generates error. $(2 \times 2 = 4)$

Your program must be written in such a way that it ensures **the maximum code reuse policy**. (2)

Write down a **main()** function to demonstrates the functionalities of your program. (2)

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